

15.2 Kinetic Theory

Question Paper

Course	CIE A Level Physics
Section	15. Ideal Gases
Topic	15.2 Kinetic Theory
Difficulty	Medium

Time allowed: 60

Score: /48

Percentage: /100

Question 1a

A cylinder contains 5.12 mol of an ideal gas at pressure 5.60×10^5 Pa and volume $3.80 \times 10^{-2} \text{ m}^3$.

Calculate the thermodynamic temperature of the gas.

temperature = K

[2 marks]

Question 1b

The average kinetic energy E_K of a molecule of the gas is given by the expression

$$E_K = \frac{3}{2} kT$$

where k is the Boltzmann constant and T is the thermodynamic temperature.

The gas is heated at constant pressure so that its temperature rises by 125 K.

(i)

Show that the new volume of the gas is $4.75 \times 10^{-2} \text{ m}^3$.

(ii)

Calculate the increase in internal energy of the gas. Explain your working.

increase in internal energy = J

[4 marks]

Question 1c

(i)

Use the information in **(b)(i)** to calculate the external work done during the expansion of the gas.

work done = J [2]

(ii)

Use the first law of thermodynamics to determine the total thermal energy transferred to the gas in **(b)**. Explain your reasoning.

energy = J [2]

[4 marks]**Question 2a**

The kinetic theory of gases equation can be written as

$$pV = \frac{1}{3}Nm\langle c^2 \rangle$$

State the meaning of each of the symbols in this equation.

[3 marks]

Question 2b

Use the equation in **(a)** to show that the average translational kinetic energy E_K of a molecule of an ideal gas is given by

$$E_K = \frac{3}{2}kT$$

[2 marks]**Question 2c**

The mass of a nitrogen molecule is 4.65×10^{-26} kg.

Use the equation in **(b)** to determine the root-mean-square (r.m.s.) speed u of a nitrogen molecule at 20°C .

Assume that nitrogen behaves as an ideal gas.

[3 marks]

Question 2d

A fixed mass of nitrogen gas at initial pressure P is sealed in a cylindrical container by a moveable piston at one end, as shown in Fig. 1.1.

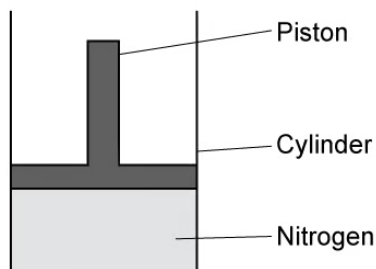


Fig. 1.1

The temperature of the gas is $20\text{ }^{\circ}\text{C}$.

At all times, the gas and the container remain in thermal equilibrium with the surroundings. The piston is slowly moved out of the cylinder so that the nitrogen gas is decompressed.

On Fig. 1.2, sketch the variation of pressure with the r.m.s. speed of the nitrogen molecules as the pressure decreases.

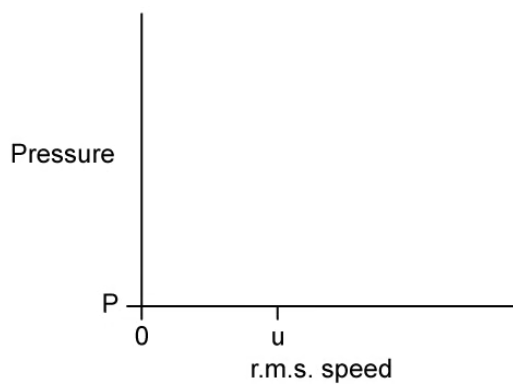


Fig. 1.2

[2 marks]

Question 3a

A fixed mass of an ideal gas is at a temperature of $24\text{ }^{\circ}\text{C}$. The volume of the gas is $3.3 \times 10^{-3}\text{ m}^3$ and the pressure is $2.1 \times 10^5\text{ Pa}$.

Calculate the number n of moles in the gas.

[2 marks]

Question 3b

The mass of one molecule of the gas is 30 u .

Determine the r.m.s speed of the gas molecules.

[3 marks]

Question 3c

The temperature of the gas is increased by $72\text{ }^{\circ}\text{C}$.

Calculate the value of the ratio

$$\frac{\text{original r.m.s. speed of molecules}}{\text{new r.m.s. speed of molecules}}$$

[2 marks]

Question 4a

State two assumptions of the simple kinetic model of a gas.

[2 marks]

Question 4b

Use the kinetic model of gases and Newton's laws of motion to explain how a gas exerts a pressure on the sides of its container.

[3 marks]

Question 4c

In a sample of gas at room temperature, four of the atoms have the following speeds:

$$1.42 \times 10^3 \text{ m s}^{-1}$$

$$1.51 \times 10^3 \text{ m s}^{-1}$$

$$1.56 \times 10^3 \text{ m s}^{-1}$$

$$1.54 \times 10^3 \text{ m s}^{-1}$$

For these four atoms, calculate the r.m.s. to three significant figures.

[4 marks]

Question 5a

A single molecule is in a cube-shaped box with sides of equal length l , as shown in Fig. 1.1.

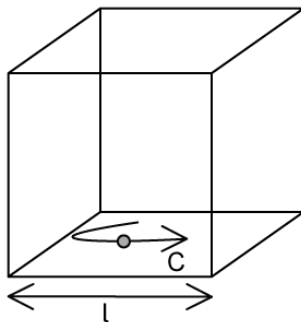


Fig. 1.1

The molecule has a mass m and moves with a speed c parallel to one side of the box. It collides at regular intervals with the ends of the box, exerting a force and contributing to the pressure of the gas.

The collisions the particle has with the container walls are **elastic**.

Determine an expression for the change in momentum of the molecule as it hits a wall perpendicularly.

[1 mark]

Question 5b

(i)

Show that the time between collisions of the molecule on the container wall is given by:

$$\text{time between collisions} = \frac{2l}{c}$$

[2]

(ii)

Determine an expression for the change in momentum per second.

[2]

[4 marks]

Question 5c

Consider N molecules identical to the single molecule in parts **(a)** and **(b)** in the same cube-shaped box.

Each molecule has a different velocity that contributes to the pressure.

Calculate the total pressure from the N particles in the box.

[4 marks]**Question 5d**

Hence, show that the kinetic theory of gases equation is:

$$pV = \frac{1}{3}Nm\langle c^2 \rangle$$

[3 marks]



Model Answers: Medium

1a

(a) Calculate the thermodynamic temperature of the gas:

List the known quantities:

- Number of mol of gas, $n = 5.12 \text{ mol}$
- Pressure of gas, $p = 5.60 \times 10^5 \text{ Pa}$
- Volume of gas, $V = 3.80 \times 10^{-2} \text{ m}^3$
- Molar gas constant, $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$

Use the ideal gas equation to determine T :

- $pV = nRT$ [1 mark]
- $T = \frac{pV}{nR} = \frac{(5.6 \times 10^5) \times (3.8 \times 10^{-2})}{5.12 \times 8.31}$
- $T = 500.15 \text{ K} = 500 \text{ K (3 s.f.)}$ [1 mark]

[Total: 2 marks]

- Using the ideal gas equation the **temperature** is given in **K**

1b

(b)

List the known quantities:

- Temperature change, $\Delta T = 125 \text{ K}$
- Initial temperature, $T_1 = 500 \text{ K}$ (from part a)
- Final temperature, $T_2 = 500 + 125 = 625 \text{ K}$
- Initial volume, $V_1 = 3.80 \times 10^{-2} \text{ m}^3$ (from part a)
- Number of moles of gas, $n = 5.12$ (from part a)

- Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
- Avogadro's constant, $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$

(i) Show that the new volume of the gas is $4.75 \times 10^{-2} \text{ m}^3$:

- $\frac{V_1}{V_2} = \frac{T_1}{T_2}$
- $V_2 = V_1 \times \frac{T_2}{T_1}$
- $V_2 = (3.8 \times 10^{-2}) \times \frac{625}{500}$
- $V_2 = 0.0475 \text{ m}^3 = 4.75 \times 10^{-2} \text{ m}^3$ [1 mark]

(ii) Calculate the increase in internal energy of the gas:

- (For an ideal gas) the change in internal energy = the change in (total) kinetic energy (of the molecules of the gas); [1 mark]

For n moles of gas with N_A number of molecules in each mole:

- $\Delta U = E_k = \frac{3}{2} NkT = \frac{3}{2} nN_A kT$
- $\Delta U = \frac{3}{2} \times 5.12 \times (6.02 \times 10^{23}) \times (1.38 \times 10^{-23}) \times 125$ [1 mark]
- $\Delta U = 7975 \text{ J} = 7980 \text{ J (3 s.f.)}$ [1 mark]

[Total: 4 marks]

- Don't let the different equations confuse you!
 - $E_K = \frac{3}{2} kT$ is the **kinetic energy** of **1 molecule** of gas
- Multiply this by the number of molecules in the gas and you get the total internal energy
 - **number of molecules (N) = number of moles (n) × number of molecules in 1 mole (N_A)**

1c

(c)

(i) Use the information in **(b)(i)** to calculate the external work done during the expansion of the gas:

List the known quantities:

- Pressure, $p = 5.60 \times 10^5 \text{ Pa}$ (from part a)
- Initial volume, $V_1 = 3.80 \times 10^{-2} \text{ m}^3$ (from part a)
- Final volume, $V_2 = 4.75 \times 10^{-2} \text{ m}^3$ (from part b)

Recall the equation for work done on a gas:

- $W = p \Delta V$ [1 mark]

Substitute the known quantities into the equation to calculate the work done, W :

- $W = p \times (V_2 - V_1)$
- $W = (5.6 \times 10^5) \times ((4.75 \times 10^{-2}) - (3.80 \times 10^{-2}))$
- $W = 5320 \text{ J}$ [1 mark]

ii) Use the first law of thermodynamics to determine the total thermal energy transferred to the gas in **(b)**. Explain your reasoning:

List the known quantities:

- Total increase in internal energy, $\Delta U = 7980 \text{ J}$ (from part b, ii)
- Work done on gas, $W = 5320 \text{ J}$ (from part c, i)

Explain the value of the work done:

- The work done by the gas is positive because the gas is expanding
OR
- Work is done by the gas; [1 mark]

Obtain an expression for the total thermal energy transferred to the gas:

- Total increase in internal energy = thermal energy transferred to gas + work done

on gas

- $\Delta U = Q + W$

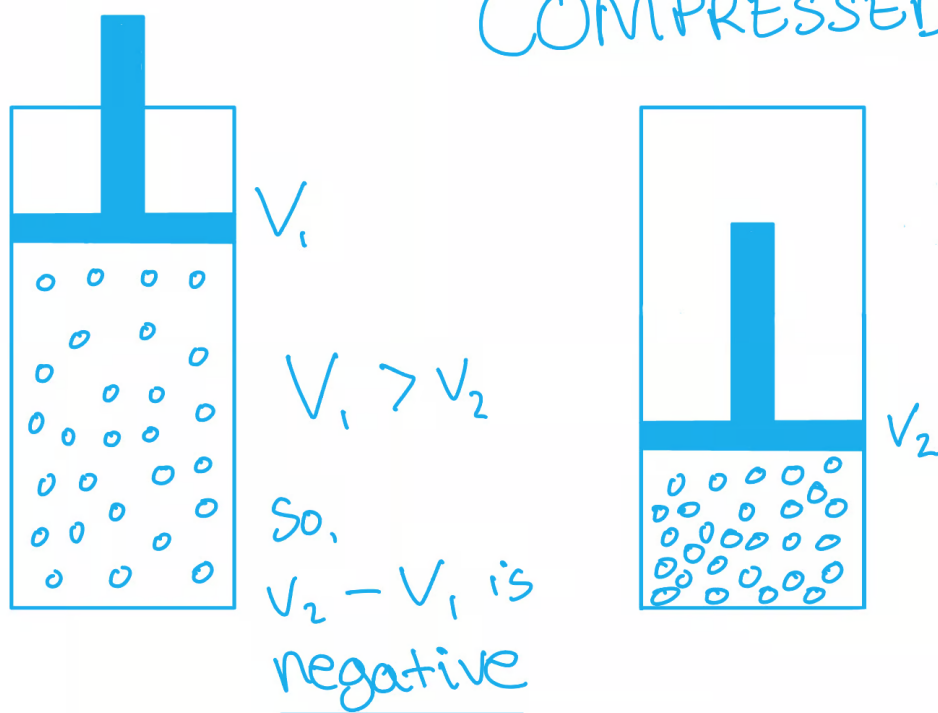
Rearrange the equation to make Q the subject and substitute the known quantities:

- $Q = \Delta U - W$
- $Q = 7980 - (-5320)$
- $Q = 13\,300 \text{ J}$ [1 mark]

[Total: 4 marks]

- It can be tricky to understand why the work done **ON** a gas is **negative**

work is done ON a gas when it is
COMPRESSED:



2a

(a) The meaning of each of the symbols in the equation is as follows:

- p = the pressure of the gas **AND** V = the volume of the gas; [1 mark]
- N = the number of molecules in the gas **AND** m = the mass of one molecule of gas; [1 mark]
- $\langle c^2 \rangle$ = the mean square speed of the molecules in the gas; [1 mark]

[Total: 3 marks]

2b

(b) Show that the average translational kinetic energy is $E_K = \frac{3}{2}kT$:

- Ideal gas equation: $pV = NkT$ and $pV = \frac{1}{3}Nm \langle c^2 \rangle$

- $\therefore NkT = \frac{1}{3}Nm \langle c^2 \rangle$ [1 mark]

- p = the pressure of the gas **AND** V = the volume of the gas; [1 mark]
- N = the number of molecules in the gas **AND** m = the mass of one molecule of gas; [1 mark]
- $\langle c^2 \rangle$ = the mean square speed of the molecules in the gas; [1 mark]

[Total: 3 marks]

2b

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- Ideal gas equation: $pV = NkT$ and $pV = \frac{1}{3}Nm\langle c^2 \rangle$
- $\therefore NkT = \frac{1}{3}Nm\langle c^2 \rangle$ [1 mark]
- $\cancel{N}kT = \frac{1}{3}\cancel{N}m\langle c^2 \rangle \Rightarrow 3kT = m\langle c^2 \rangle$
- Divide both sides by 2: $\frac{3}{2}kT = \frac{1}{2}m\langle c^2 \rangle$ [1 mark]
- $E_K = \frac{1}{2}m\langle c^2 \rangle$ therefore, $E_K = \frac{3}{2}kT$

[Total: 2 marks]

The expression gives the **average** translational kinetic energy, so the expression $3kT = m\langle c^2 \rangle$ must be $\div 2$

The kinetic energy equation from mechanics = $\frac{1}{2}mv^2$ where $v^2 = \langle c^2 \rangle$

Hence, $E_k = \frac{3}{2}kT = \frac{1}{2}m\langle c^2 \rangle$

2c

(c) Determine the root-mean-square (r.m.s.) speed u of a nitrogen molecule at 20 °C:

List the known quantities:

- Mass of nitrogen molecule, $m = 4.65 \times 10^{-26} \text{ kg}$
- Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
- Temperature, $T = 20^\circ\text{C} = 20 + 273 = 293 \text{ K}$ [1 mark]

Rearrange and substitute into the expression from **(b)** to find u :

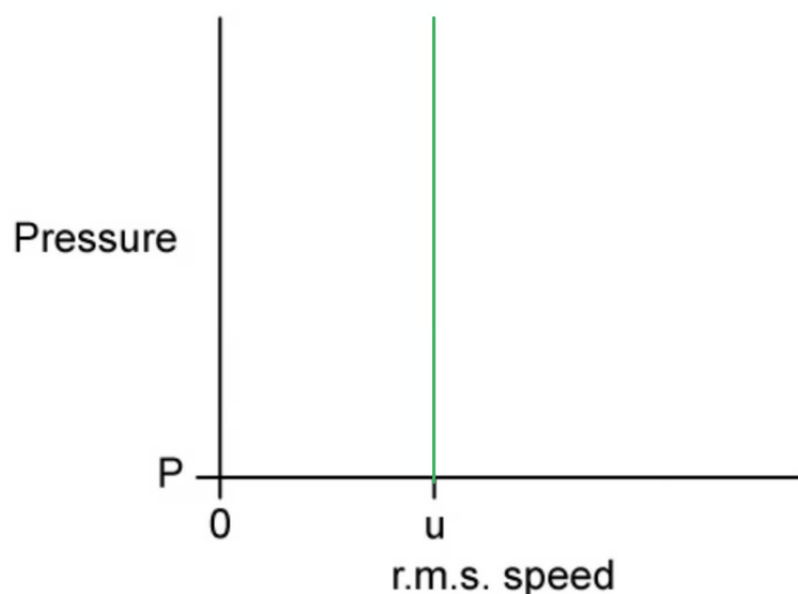
- $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \Rightarrow mu^2 = 3kT$
- $mu^2 = 3kT \Rightarrow u = \sqrt{\frac{3kT}{m}}$
- $u = \sqrt{\frac{3 \times (1.38 \times 10^{-23}) \times 293}{(4.65 \times 10^{-26})}}$ [1 mark]
- $u = 510.75 \text{ m s}^{-1} = 511 \text{ m s}^{-1} \text{ (3 s.f.)}$ [1 mark]

[Total: 3 marks]

When making any calculations to do with ideal gases, always convert **temperature** into **K**.

2d

(d) The graph of pressure with the r.m.s. speed is as follows:



- A straight line passing through (u, P) ; [1 mark]
- The line is vertical; [1 mark]

[Total: 2 marks]

$E_K \propto T$ so E_K is independent of P . Hence the r.m.s. remains constant as the pressure decreases.

3a

(a) Calculate the number n of moles in the gas:

List the known quantities:

- Temperature of gas, $T = 24^\circ\text{C} = 24 + 273 = 297\text{ K}$
- Volume of gas, $V = 3.3 \times 10^{-3}\text{ m}^3$
- Pressure of gas, $P = 2.1 \times 10^5\text{ Pa}$
- Molar gas constant, $R = 8.31\text{ J K}^{-1}\text{ mol}^{-1}$

State the ideal gas equation, rearrange and substitute the known quantities:

- $pV = nRT \Rightarrow n = \frac{pV}{RT}$
- $n = \frac{(2.1 \times 10^5) \times (3.3 \times 10^{-3})}{8.31 \times 297}$ [1 mark]
- $n = 0.28$ [1 mark]

[Total: 2 marks]

3b

(b) Determine the r.m.s speed of the gas molecules:

List the known quantities:

- Temperature of gas, $T = 297\text{ K}$
- Mass of one molecule, $m = 30\text{ u} = 30 \times (1.66 \times 10^{-27}) = 4.98 \times 10^{-26}\text{ kg}$
- Boltzmann constant, $k = 1.38 \times 10^{-23}\text{ J K}^{-1}$

Use the equation for the average kinetic energy of one molecule of a gas:

- $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$ [1 mark]
- $c = \sqrt{\frac{3kT}{m}}$
- $c = \sqrt{\frac{3 \times (1.38 \times 10^{-23}) \times 297}{(4.98 \times 10^{-26})}}$ [1 mark]
- $c = 496.9 \text{ m s}^{-1} = 497 \text{ m s}^{-1}$ (3 s.f.) [1 mark]

[Total: 3 marks]

Remember to convert the mass to **kg**. There are **many equations** in this topic. **Learn** the equations given for 1 molecule and how it will change for 1 mole / a certain number of particles.

The other longer way to complete this question is to consider the **number of particles**:

$$\text{using } N = nN_A \text{ and } pV = \frac{1}{3}Nm\langle c^2 \rangle$$

3c

(c) Calculate the ratio of original to new r.m.s. speed:

List the known quantities:

- Original temperature, $T_1 = 297 \text{ K}$
- Change in temperature = $72^\circ\text{C} = 72 \text{ K}$
- New temperature, $T_2 = 297 + 72 = 369 \text{ K}$
- Original r.m.s. speed of molecules = c_1
- New r.m.s. speed of molecules = c_2

Substitute the known quantities into the ratio:

- r.m.s. speed of molecules: $c \propto \sqrt{T}$ [1 mark]

- $\frac{c_1}{c_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{297}{369}}$
- $\frac{c_1}{c_2} = 0.8971 = 0.897 \text{ (3 s.f.) [1 mark]}$

[Total: 2 marks]

You can answer this question by **calculating** the **r.m.s speed** of one molecule of gas at the new temperature using the method from (c). This will give you the same correct answer but will take a lot **longer**!

- $\frac{c_1}{c_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{297}{369}}$
- $\frac{c_1}{c_2} = 0.8971 = 0.897 \text{ (3 s.f.) [1 mark]}$

[Total: 2 marks]

You can answer this question by **calculating** the **r.m.s speed** of one molecule of gas at the new temperature using the method from (c). This will give you the same correct answer but will take a lot **longer**!

Both the **K** and **°C** temperature scales are **linear**, so an increase in 72 °C = an increase in 72 K.

4a

(a) Assumptions of the simple kinetic model of a gas:

Any **two** from:

- The gas contains a large number of molecules / atoms / particles; [1 mark]
- The molecules are in random motion; [1 mark]
- There are no intermolecular forces between the molecules; [1 mark]
- The collisions between the molecules are elastic; [1 mark]
- The time of the collisions is much less than the time between the collisions; [1 mark]
- The volume of molecules is much less than the volume of the containing vessel; [1 mark]

[Total: 2 marks]

4b

(b) A gas exerts a pressure on the sides of its container because:

Any **three** from:

- The molecules collide with the walls; [1 mark]
- The change in momentum of the molecules (on collision with the walls) means

there is a force (on the molecules); [1 mark]

- The molecules exert an equal and opposite force on the walls (of the container); [1 mark]
- Pressure is the average of the force from the collisions per surface area on the walls; [1 mark]

[Total: 3 marks]

4c

(c) Calculate the r.m.s. of the four speeds:

- Mean square speed: $\langle c^2 \rangle = \frac{\text{sum of (4 values)}^2}{4}$
- $\langle c^2 \rangle = \frac{(1420)^2 + (1510)^2 + (1560)^2 + (1540)^2}{4}$ [1 mark]
- $\langle c^2 \rangle = 2.275 \times 10^6 \text{ m}^2 \text{ s}^{-2}$ [1 mark]

Calculate the r.m.s. speed:

- $r.m.s. = \sqrt{\langle c^2 \rangle}$
- $r.m.s. = \sqrt{2.275 \times 10^6}$ [1 mark]
- $r.m.s. = 1.51 \times 10^3 \text{ m s}^{-1}$ (3 s.f.) [1 mark]

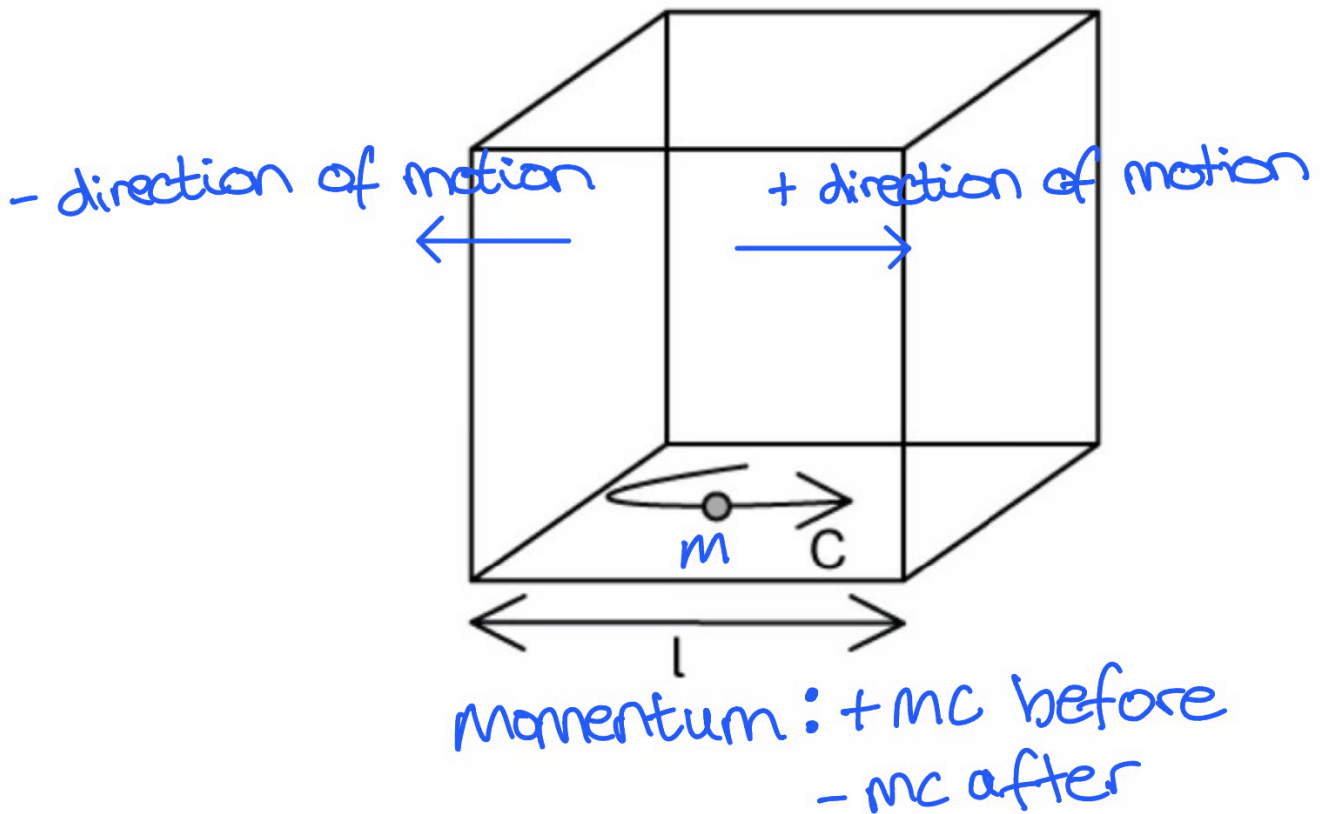
[Total: 4 marks]

Finding the **mean speed** $\neq$ finding the **root mean square speed**. Be sure to **show all of the steps** above in your answer to gain full marks.

5a

(a) Consider the change in momentum of the molecule as it hits the wall perpendicularly:

- Change in momentum = momentum after – momentum before
- $\Delta p = -mc - (+mc)$
- $\Delta p = -2mc$ [1 mark]

[Total: 1 mark]

5b

(b)

(i) Write an expression for the time between collisions with the wall:

- The molecule travels from one side of the cube to the surface it collides with, this is a distance l
- After the collision, the molecule travels back across one side of the cube, the same distance l
- Hence, the cube travels a total distance of $2l$ [1 mark]

Obtain an expression for the time between collisions:

- $time = \frac{distance}{speed}$

- $\text{time between collisions} = \frac{\text{distance between collisions}}{\text{speed of molecule}} = \frac{2l}{c}$ [1 mark]

(ii) Write an expression for the rate of change in momentum:

- Rate of change of momentum: $\frac{\text{change in momentum}}{\text{time between collisions}} = \frac{\Delta p}{\Delta t}$ [1 mark]

- Rate of change of momentum: $\frac{\Delta p}{\Delta t} = \frac{2mc}{\left(\frac{2l}{c}\right)} = \frac{mc^2}{l}$ [1 mark]

[Total: 4 marks]

5c

(c) The total pressure from the N particles in the box is:

- All the collisions take place on one face of the cube
- Area of the collision: $l \times l = l^2$ [1 mark]

Obtain an expression for pressure and substitute for force and area:

- Force = rate of change of momentum

- $\text{pressure} = \frac{\text{force}}{\text{area}} = \frac{\left(\frac{mc^2}{l}\right)}{l^2}$ [1 mark]

Obtain an expression that considers the different velocities of the molecules:

- Total pressure of the molecules: $p = \frac{Nmc^2}{l^3}$ [1 mark]
- The different velocities can be described by the mean square of the velocities $\langle c^2 \rangle$

- Hence, $p = \frac{Nm \langle c^2 \rangle}{l^3}$ [1 mark]

[Total: 4 marks]

5d

(d) Show that the kinetic theory of gases equation is $pV = \frac{1}{3}Nm \langle c^2 \rangle$:

- Velocity components in 3 dimensions: $c^2 = (c_x)^2 + (c_y)^2 + (c_z)^2$

- Hence, $p = \frac{Nm \langle c^2 \rangle}{l^3}$ [1 mark]

[Total: 4 marks]

5d

(d) Show that the kinetic theory of gases equation is $pV = \frac{1}{3}Nm \langle c^2 \rangle$:

- Velocity components in 3 dimensions: $c^2 = (c_x)^2 + (c_y)^2 + (c_z)^2$
- Because of symmetry, it can be assumed that $\langle (c_x)^2 \rangle = \langle (c_y)^2 \rangle = \langle (c_z)^2 \rangle$
- Mean square speed: $\langle (c_x)^2 \rangle = \frac{1}{3} \langle c^2 \rangle$ [1 mark]

Consider the volume of the cube:

- The cube is an $l \times l \times l$ box
- Hence, volume $V = l^3$ [1 mark]

Substitute the values into the expression for pressure:

- $p = \frac{Nm \langle c^2 \rangle}{l^3}$
- $p = \frac{1}{3} \times \frac{Nm \langle c^2 \rangle}{V}$ [1 mark]
- Hence, $pV = \frac{1}{3}Nm \langle c^2 \rangle$

[Total: 3 marks]

It can be tricky to think about the **different components of velocity**:

